FNCE30007 DERIVATIVE SECURITIES

Tutorial 6 solutions

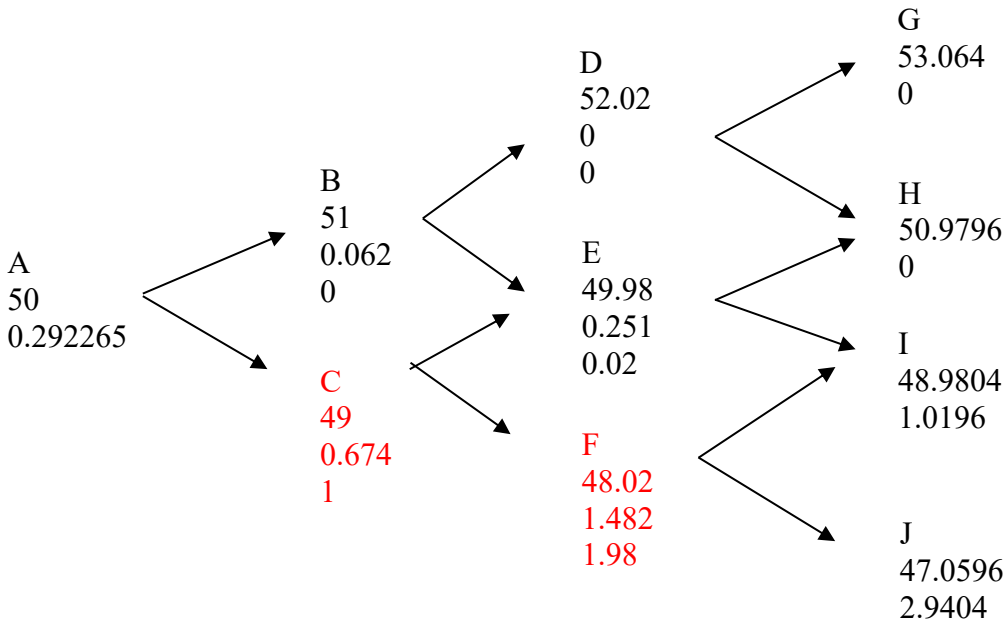
Binomial II

QUESTION 1

Consider a 3-month American put option on a non dividend paying stock when the stock price is \$50, the strike price is \$50, the risk-free rate of interest is 12% p.a. and the underlying stock price is expected to increase or decrease by 2% per month.

(i) Use a three-period model to calculate the current value of the option.

(ii) Should the option be exercised early? Check your answer using the simulator.



$$p = \frac{e^{r\Delta t} - d}{u - d}$$

$$= \frac{1.01005 - 0.98}{1.02 - 0.98}$$

$$= 0.751$$

where $\Delta t = \frac{1}{12}, u = 1.02, d = 0.98$

Use $f = e^{-rT}[p * fu + (1 - p)fd]$ to find the option value at each node from A to F

For example, to get the option payoff at F

$$f = [0.751 * 1.0196 + (1 - 0.751) * 2.9404]/1.01005$$

$$= 1.482$$

Compare this value with the intrinsic value with early exercise (K-S_T). If intrinsic value is higher, then exercise early at that node.

Based on the calculation, one should exercise early if stock price drops from $50 to $49 at node C.

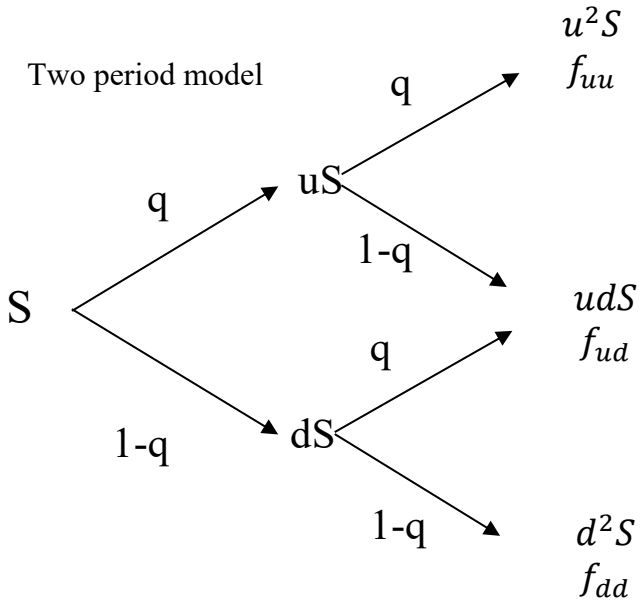
See Excel for the detailed calculation at each node.

QUESTION 2

Consider a two-period binomial model. Let $p = \frac{e^{r\Delta t} - d}{u - d}$ be the probability of a stock price increase in a risk neutral world. Use the principle of Risk Neutral valuation to derive the two period (European) valuation formula:

$$f = \frac{p^2 f_{uu} + 2p(1-p) f_{ud} + (1-p)^2 f_{dd}}{e^{2r\Delta t}}$$

Two period model



q represents for the real-world probabilities of up/down movements

In the risk neutral portfolio, we use risk-neutral probability such that $p = \frac{e^{r\Delta t} - d}{u - d}$ instead of q

From the tree diagram, we can tell $E^*(f_T) = p^2 f_{uu} + 2p(1-p) f_{ud} + (1-p)^2 f_{dd}$

By Risk Neutral Valuation:

$$f = e^{-rT} E^*(f_T)$$

$$f = \frac{p^2 f_{uu} + 2p(1-p) f_{ud} + (1-p)^2 f_{dd}}{e^{2r\Delta t}}$$

where T = $2\Delta t$

Futures Options

QUESTION 3

Using two portfolios, derive the lower bound for the price of a European futures put option. In doing this, detail the composition of each portfolio and draw a table illustrating the initial and terminal values of the portfolios.

$p \geq \max [e^{-rT}(K - F_0), 0]$
To prove $p \geq e^{-rT}(K - F_0)$, we need to set up two portfolios, A and B.
Portfolio A includes a cash amount equal to the PV of K (= $K*e^{-rT}$).
Portfolio B includes
a long put futures option;
a long futures contract; and
a cash amount equal to the PV of F_0 (= F_0*e^{-rT}).

	t = 0	t = T	
		$F_T > K$	$F_T \leq K$
A	$K*e^{-rT}$	K	K
Net portfolio value	$K*e^{-rT}$	K	K
B	p	0	$K - F_T$
	0	$F_T - F_0$	$F_T - F_0$
	F_0*e^{-rT}	F_0	F_0
Net portfolio value	$p + F_0*e^{-rT}$	F_T	K

At time T,
if $F_T \leq K$, then A=B;
if $F_T > K$, then B>A.
So at time T, we have $B \geq A$.

At time t_0, the values of A and B must follow the same relation, and therefore we have
$p + F_0*e^{-rT} \geq K*e^{-rT}$
$p \geq (K - F_0)* e^{-rT}$

Because p can never have a negative value, $p \geq \max [e^{-rT}(K - F_0), 0]$.

QUESTION 4

Suppose you buy a put option contract on October gold futures with a strike price of $900 per ounce. Each contract is for the delivery of 100 ounces. What happens if you exercise when the October futures price is $880?

An amount $(900 - 880) \times 100 = \$2{,}000$ is added to your margin account and you acquire a short futures position obligating you to sell 100 ounces of gold in October. This position is marked to market in the usual way until you choose to close it out.

QUESTION 5

Suppose that a futures price is currently 30. The risk-free interest rate is 5% per annum. A three-month American call futures option with a strike price of 28 is worth 4. Calculate bounds for the price of a three-month American put futures option with a strike price of 28.

$C - P$ must lie between

$$30e^{-0.05\times 3/12} - 28 = 1.63$$

and

$$30 - 28e^{-0.05\times 3/12} = 2.35$$

Because $C = 4$ we must have $1.63 < 4 - P < 2.35$ or

$$1.65 < P < 2.37$$

QUESTION 6

After reading the Virgin case, complete the following questions before the tutorial.

All market data are as at 2 September 2026.

QUESTION DEFERRED TO NEXT TUTORIAL - SOLUTION WILL BE RELEASED NEXT WEEK!